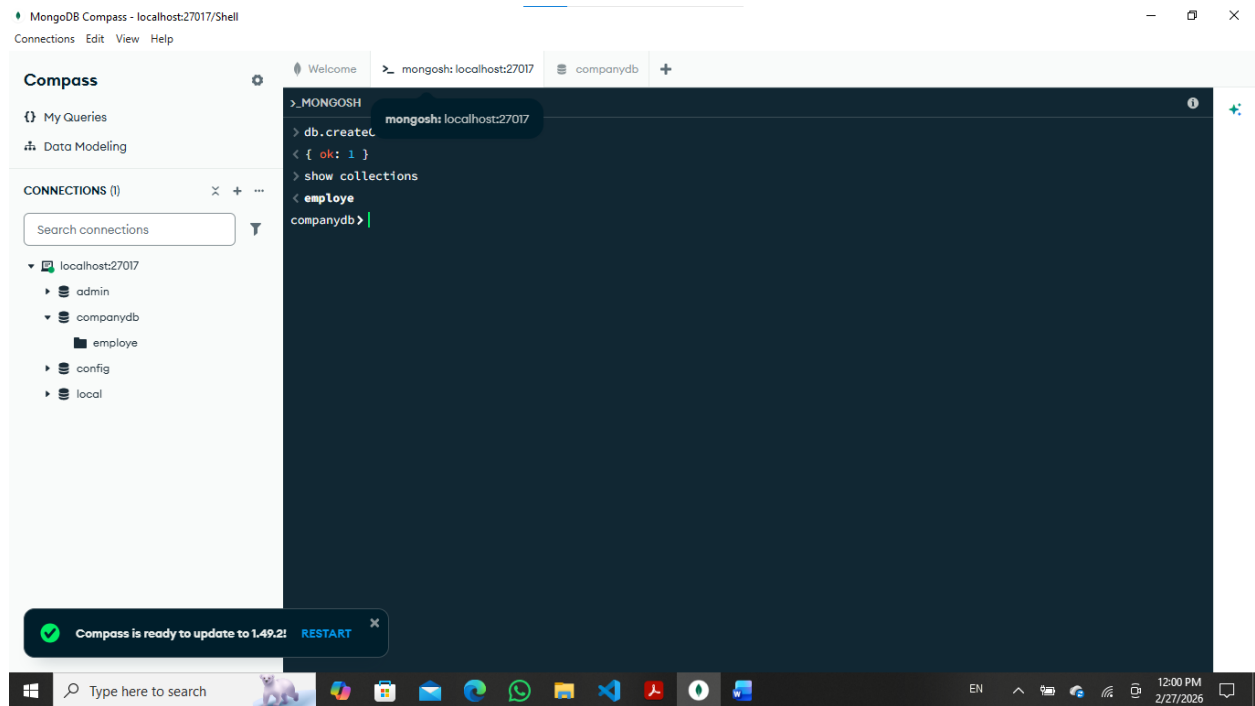


New database >collection



Task 1: Insert Documents

MongoDB Compass - localhost:27017/Shell

Connections Edit View Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

localhost:27017

- admin
- companydb
 - employee
- config
- local

employee companydb mongosh: localhost:27017

```
> use companydb
< switched to db companydb
companydb> db.companydb.insertOne({
  "eid" : "002",
  "name" : "sumedh jadhav",
  "age" : "23",
  "department" : "IT",
  "designation" : "software engineer",
  "salary" : "55000",
  "city" : "karjat",
  "joining_date" : "05-02-2025"
})
```

12:38 PM 2/27/2026

MongoDB Compass - localhost:27017/companydb.employees

Connections Edit View Collection Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

localhost:27017

- admin
- companydb
 - employees
- config
- local

companydb mongosh: localhost:27017 employees

localhost:27017 > companydb > employees

Documents 10 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain Reset Find Options

ADD DATA UPDATE DELETE EXPORT DATA EXPORT CODE 25 1 - 10 of 10

joining_date: 2023-08-20T00:00:00.000+00:00

Document 1

```
{
  "_id": ObjectId('69a14630d4aece84a622d796'),
  "eid": "005",
  "name": "Rahul Patil",
  "age": 30,
  "department": "Marketing",
  "designation": "Marketing Manager",
  "salary": 60000,
  "city": "Nashik",
  "joining_date": 2022-06-10T00:00:00.000+00:00
}
```

Document 2

```
{
  "_id": ObjectId('69a14630d4aece84a622d797'),
  "eid": "006",
  "name": "Sneha Kulkarni",
  "age": 24,
  "department": "IT",
  "designation": "Frontend Developer",
  "salary": 52000,
  "city": "Thane",
  "joining_date": 2025-01-18T00:00:00.000+00:00
}
```

12:55 PM 2/27/2026

Task 2: Basic Queries

find data

The screenshot shows the MongoDB Compass interface. The left sidebar displays the 'Connections' panel with a search bar and a list of connections. The main panel shows a query in the 'MONGOSH' shell. The query is:

```
> use companydb
< switched to db companydb
> db.employees.find()
< {
  _id: ObjectId('69a145c0d4aece84a622d78e1'),
  eid: '002',
  name: 'Sumedh Jadhav',
  age: 23,
  department: 'IT',
  designation: 'Software Engineer',
  salary: 55000,
  city: 'Karjat',
  joining_date: 2025-02-05T00:00:00.000Z
}
{
  _id: ObjectId('69a145c0d4aece84a622d78f1'),
  eid: '003',
  name: 'Amit Sharma',
  age: 28,
  department: 'HR',
  designation: 'HR Executive',
  salary: 45000,
  city: 'Pune',
  joining_date: 2024-11-15T00:00:00.000Z
}
```

The bottom of the image shows the Windows taskbar with the search bar and various application icons. The system clock indicates 12:57 PM on 2/27/2026.

Find employees who work in the "IT" department.

MongoDB Compass - localhost:27017/Shell

Connections Edit View Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

localhost:27017

- admin
- companydb
 - employees
- config
- local

companydb mongosh: localhost:27017 employees mongosh: localhost:27017

```
>_MONGOSH
age: 24,
department: 'IT',
designation: 'Frontend Developer',
salary: 52000,
city: 'Thane',
joining_date: 2025-01-18T00:00:00.000Z
}
> db.employees.find({department : "IT"})
< {
  _id: ObjectId('69a145c0d4aece84a622d78e'),
  eid: '002',
  name: 'Sumedh Jadhav',
  age: 23,
  department: 'IT',
  designation: 'Software Engineer',
  salary: 55000,
  city: 'Karjat',
  joining_date: 2025-02-05T00:00:00.000Z
}
{
  _id: ObjectId('69a145c0d4aece84a622d792'),
  eid: '006',
  name: 'Sneha Kulkarni',
  age: 24,
  department: 'IT',
```

Windows taskbar: Type here to search, 1:00 PM 2/27/2026

find salary greater than 50000

MongoDB Compass - localhost:27017/Shell

Connections Edit View Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

localhost:27017

- admin
- companydb
 - employees
- config
- local

```
>_MONGOSH
>joining_date: 2025-01-18T00:00:00.000Z
}
}
>db.employees.find({ salary: { $gt: 50000 } })
< {
  _id: ObjectId('69a145c0d4aece84a622d78e'),
  eid: '002',
  name: 'Sumedh Jadhav',
  age: 23,
  department: 'IT',
  designation: 'Software Engineer',
  salary: 55000,
  city: 'Karjat',
  joining_date: 2025-02-05T00:00:00.000Z
}
{
  _id: ObjectId('69a145c0d4aece84a622d791'),
  eid: '005',
  name: 'Rahul Patil',
  age: 30,
  department: 'Marketing',
  designation: 'Marketing Manager',
  salary: 60000,
  city: 'Nashik',
  joining_date: 2022-06-16T00:00:00.000Z
}
}
```

Windows taskbar: Type here to search, 1:01 PM 2/27/2026

Find one employee from "HR" department.

MongoDB Compass - localhost:27017/Shell

Connections Edit View Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

localhost:27017

- admin
- companydb
 - employees
- config
- local

```
>_MONGOSH
}
{
  _id: ObjectId('69a14630d4aece84a622d797'),
  eid: '006',
  name: 'Sneha Kulkarni',
  age: 24,
  department: 'IT',
  designation: 'Frontend Developer',
  salary: 52000,
  city: 'Thane',
  joining_date: 2025-01-18T00:00:00.000Z
}
>db.employees.findOne({ department: "HR" })
< {
  _id: ObjectId('69a145c0d4aece84a622d78f1'),
  eid: '003',
  name: 'Amit Sharma',
  age: 28,
  department: 'HR',
  designation: 'HR Executive',
  salary: 45000,
  city: 'Pune',
  joining_date: 2024-11-15T00:00:00.000Z
}
companydb>
```

Windows taskbar: Type here to search, 1:03 PM 2/27/2026

Task 3: Updates

Increase salary of "Yash Diwate" by 10%.

The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists 'localhost:27017' with databases 'admin', 'companydb', and 'local'. The 'companydb' database is selected, and the 'employees' collection is visible. The main panel shows the MongoDB Shell with the following commands and results:

```
> db.employees.findOne({ department: "HR" })
< {
  _id: ObjectId('69a145c0d4aece84a622d78f'),
  eid: '003',
  name: 'Amit Sharma',
  age: 28,
  department: 'HR',
  designation: 'HR Executive',
  salary: 45000,
  city: 'Pune',
  joining_date: 2024-11-15T00:00:00.000Z
}

> db.employees.updateOne(
  { name: "Yash Diwate" },
  { $mul: { salary: 1.10 } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
```

update sumedh designation

The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists 'localhost:27017' with databases 'admin', 'companydb', and 'local'. The 'companydb' database is selected, and the 'employees' collection is visible. The main panel shows the MongoDB Shell with the following commands and results:

```
> db.employees.updateOne(
  { name: "Sumedh Jadhav" },
  { $set: { designation: "Senior Developer" } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}

> db.employees.find({ designation : "Senior Developer"})
< {
  _id: ObjectId('69a145c0d4aece84a622d78e'),
  eid: '002',
  name: 'Sumedh Jadhav',
  age: 23,
  department: 'IT',
  designation: 'Senior Developer',
  salary: 55000,
  city: 'Karjat',
  joining_date: 2025-02-05T00:00:00.000Z
}
```

Change the city of all employees in "Nagpur" to "Pune".

MongoDB Compass - localhost:27017/Shell

Connections Edit View Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

localhost:27017

- admin
- companydb
 - employees
- config
- local

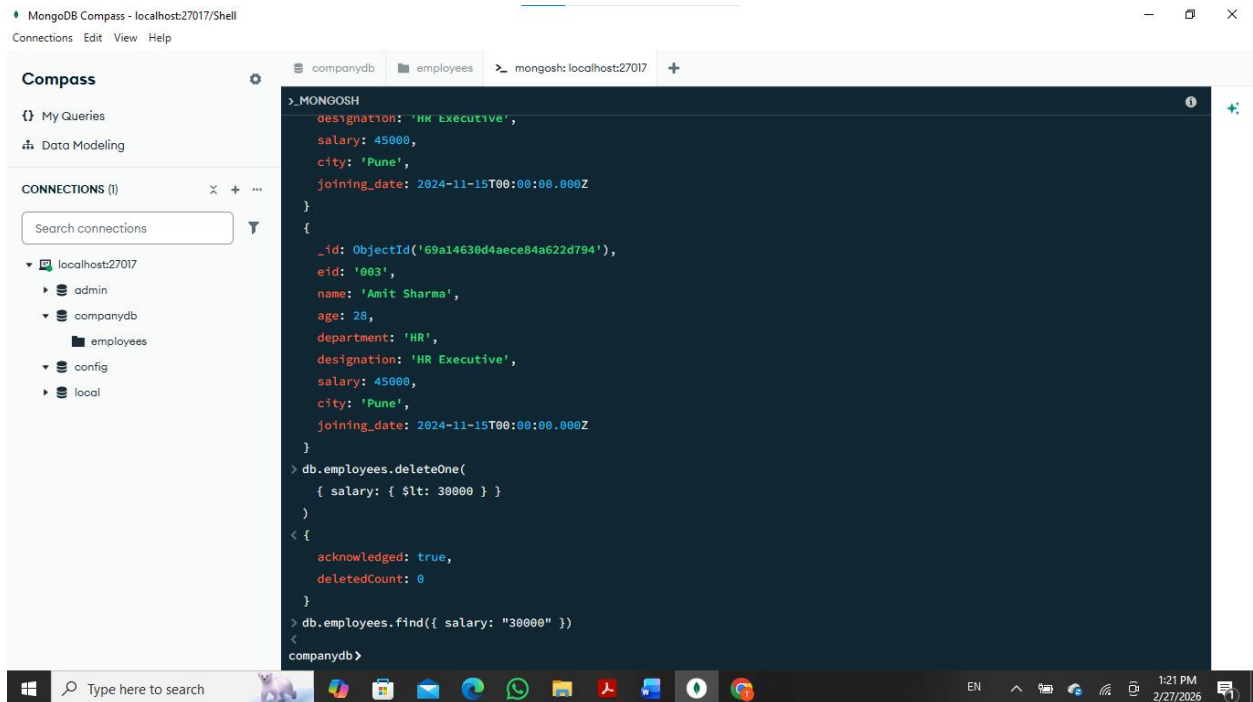
companydb employees mongosh localhost:27017

```
>_MONGOSH
> db.employees.find({city : "pune"})
<
> db.employees.find({city : "pune"})
<
> db.employees.find({ city : "pune"})
<
> db.employees.find({ city: "Pune" })
<
{
  _id: ObjectId('69a145c0d4aece84a622d78f'),
  eid: '003',
  name: 'Amit Sharma',
  age: 28,
  department: 'HR',
  designation: 'HR Executive',
  salary: 45000,
  city: 'Pune',
  joining_date: 2024-11-15T00:00:00.000Z
}
{
  _id: ObjectId('69a14630d4aece84a622d794'),
  eid: '003',
  name: 'Amit Sharma',
  age: 28,
  department: 'HR',
  designation: 'HR Executive',
  salary: 45000,
  city: 'Pune',
  joining_date: 2024-11-15T00:00:00.000Z
}
```

Windows taskbar: Type here to search, 1:17 PM, 2/27/2026

Task 4: Deletions

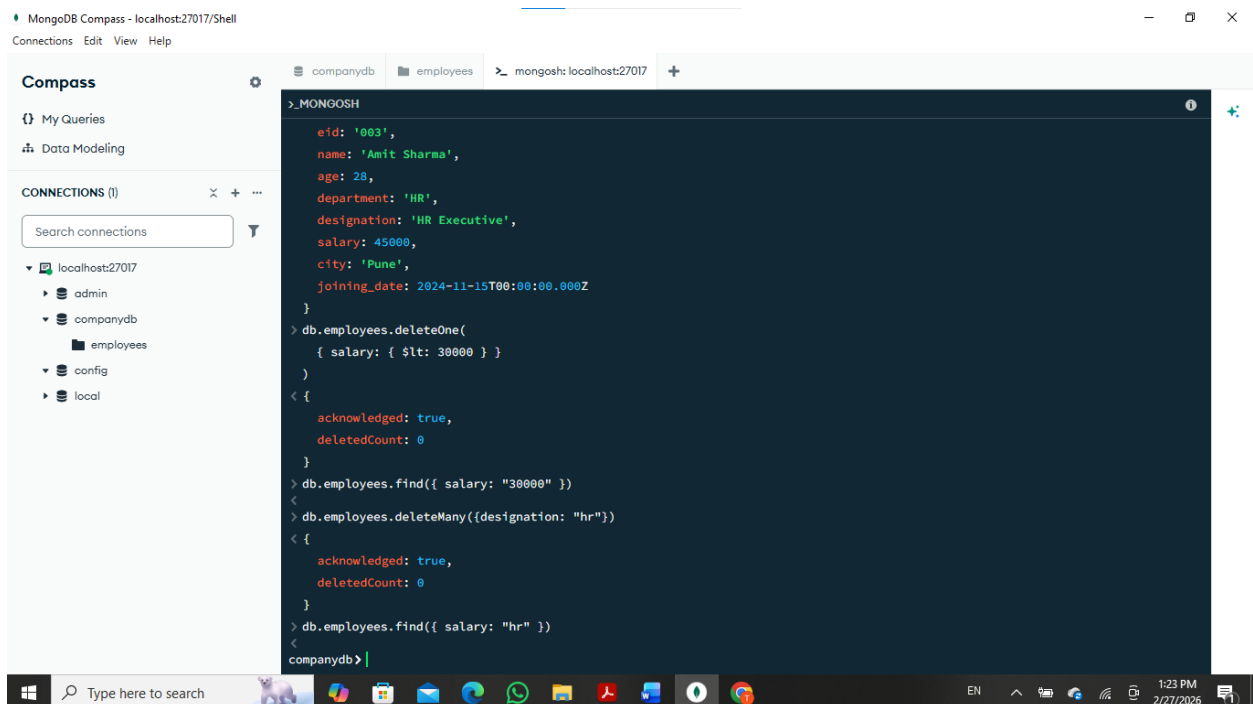
Delete one employee with salary less than 30,000.



The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel shows a connection to 'localhost:27017' with a database named 'companydb' and a collection named 'employees'. The main panel displays a JSON document for an employee with the following fields: `designation: 'HR Executive', salary: 45000, city: 'Pune', joining_date: 2024-11-15T00:00:00.000Z`. Below the document, the command `db.employees.deleteOne({ salary: { $lt: 30000 } })` is entered in the shell. The output shows `acknowledged: true, deletedCount: 0`. The bottom status bar shows the time as 1:21 PM on 2/27/2026.

```
>_MONGOOSH
{
  designation: 'HR Executive',
  salary: 45000,
  city: 'Pune',
  joining_date: 2024-11-15T00:00:00.000Z
}
{
  _id: ObjectId('69a14630d4aece84a622d794'),
  eid: '003',
  name: 'Amit Sharma',
  age: 28,
  department: 'HR',
  designation: 'HR Executive',
  salary: 45000,
  city: 'Pune',
  joining_date: 2024-11-15T00:00:00.000Z
}
> db.employees.deleteOne(
  { salary: { $lt: 30000 } }
)
< {
  acknowledged: true,
  deletedCount: 0
}
> db.employees.find({ salary: "30000" })
<
companydb>
```

Delete all employees from "Intern/hr" designation.

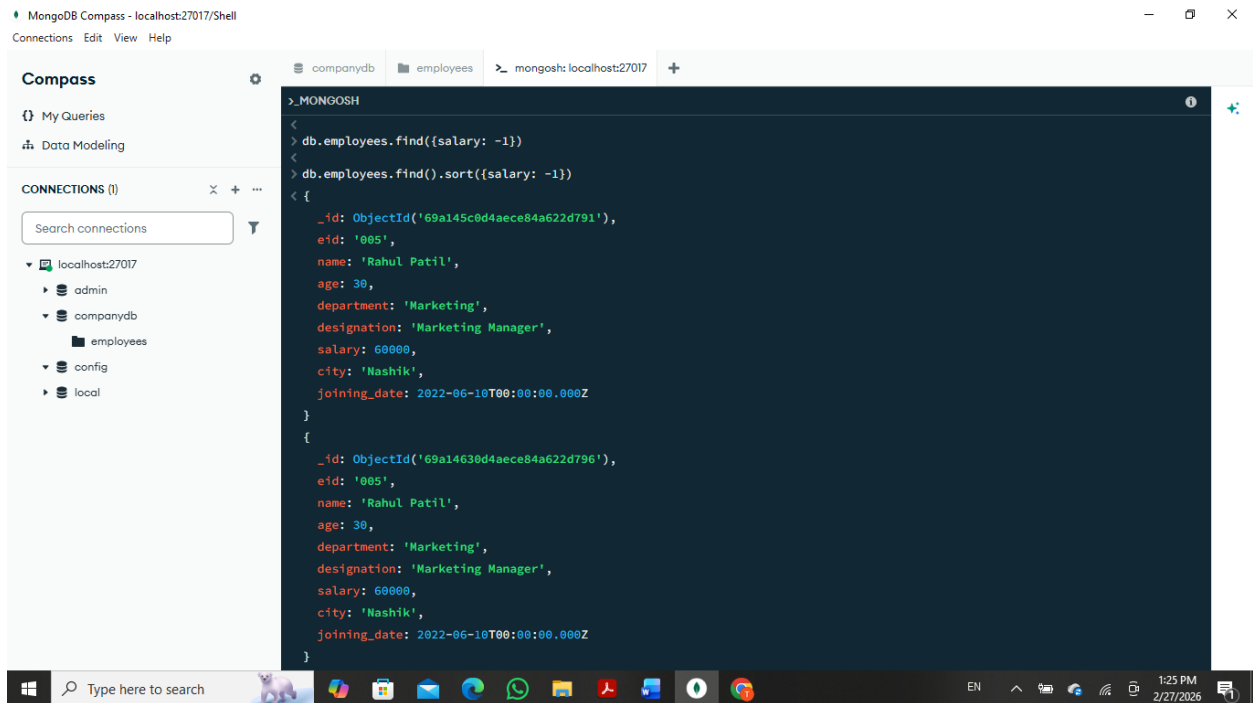


The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel shows a connection to 'localhost:27017' with a database named 'companydb' and a collection named 'employees'. The main panel displays a JSON document for an employee with the following fields: `eid: '003', name: 'Amit Sharma', age: 28, department: 'HR', designation: 'HR Executive', salary: 45000, city: 'Pune', joining_date: 2024-11-15T00:00:00.000Z`. Below the document, the command `db.employees.deleteMany({ designation: "hr" })` is entered in the shell. The output shows `acknowledged: true, deletedCount: 0`. The bottom status bar shows the time as 1:23 PM on 2/27/2026.

```
>_MONGOOSH
{
  eid: '003',
  name: 'Amit Sharma',
  age: 28,
  department: 'HR',
  designation: 'HR Executive',
  salary: 45000,
  city: 'Pune',
  joining_date: 2024-11-15T00:00:00.000Z
}
> db.employees.deleteOne(
  { salary: { $lt: 30000 } }
)
< {
  acknowledged: true,
  deletedCount: 0
}
> db.employees.find({ salary: "30000" })
<
> db.employees.deleteMany({ designation: "hr" })
< {
  acknowledged: true,
  deletedCount: 0
}
> db.employees.find({ salary: "hr" })
<
companydb>
```


Task 5: Advanced Queries

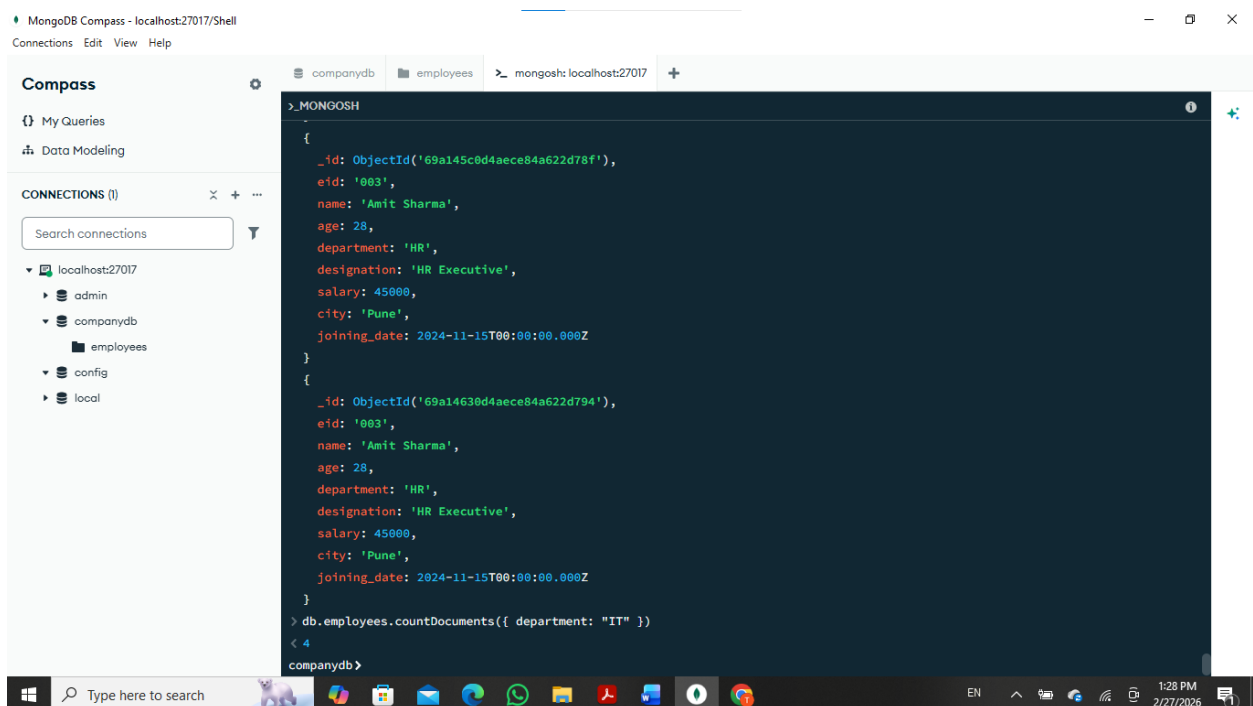
Find employees sorted by salary in descending order.



The screenshot shows the MongoDB Compass interface. The left sidebar displays the 'CONNECTIONS' list with 'localhost:27017' selected, showing databases like 'companydb' and 'employees'. The main panel shows the MongoDB Shell with the following commands and results:

```
>_MONGOSH
<
> db.employees.find({salary: -1})
<
> db.employees.find().sort({salary: -1})
< {
  "_id": ObjectId('69a145c0d4aece84a622d791'),
  "eid": '005',
  "name": 'Rahul Patil',
  "age": 30,
  "department": 'Marketing',
  "designation": 'Marketing Manager',
  "salary": 69000,
  "city": 'Nashik',
  "joining_date": 2022-06-16T00:00:00.000Z
}
{
  "_id": ObjectId('69a14630d4aece84a622d796'),
  "eid": '005',
  "name": 'Rahul Patil',
  "age": 30,
  "department": 'Marketing',
  "designation": 'Marketing Manager',
  "salary": 69000,
  "city": 'Nashik',
  "joining_date": 2022-06-16T00:00:00.000Z
}
```

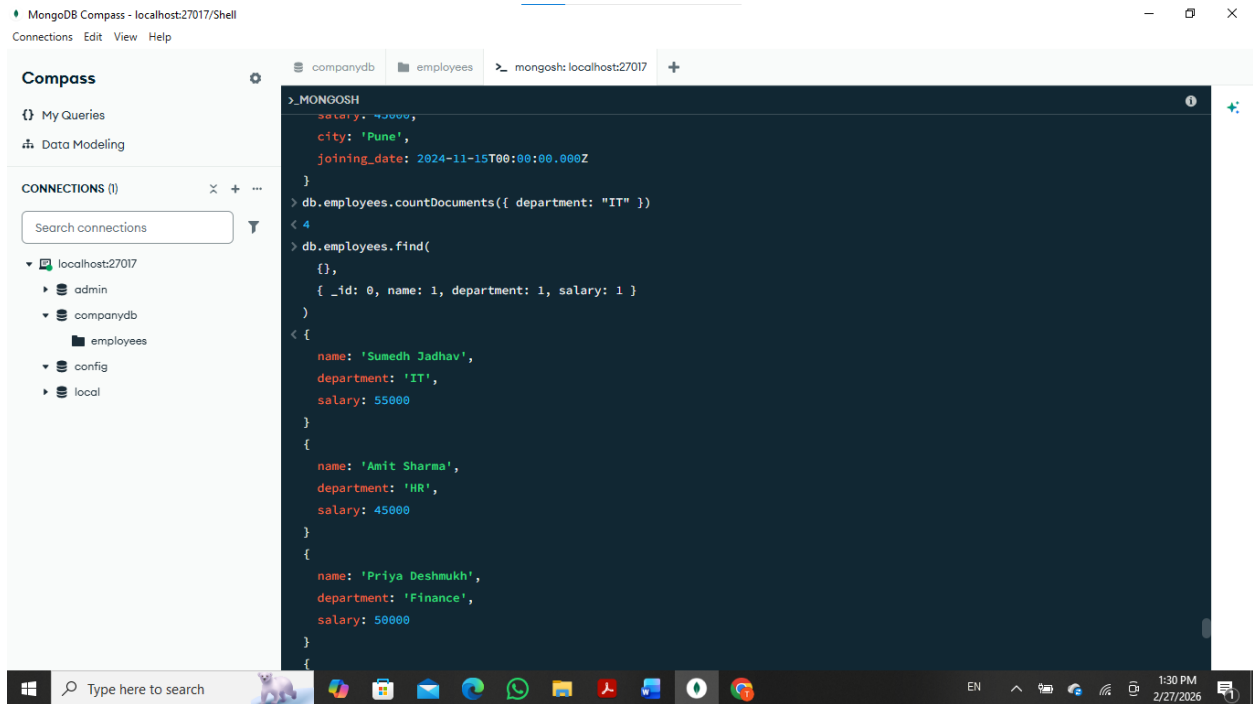
Count how many employees work in "IT" department



The screenshot shows the MongoDB Compass interface. The left sidebar displays the 'CONNECTIONS' list with 'localhost:27017' selected, showing databases like 'companydb' and 'employees'. The main panel shows the MongoDB Shell with the following commands and results:

```
>_MONGOSH
{
  "_id": ObjectId('69a145c0d4aece84a622d78f'),
  "eid": '003',
  "name": 'Amit Sharma',
  "age": 28,
  "department": 'HR',
  "designation": 'HR Executive',
  "salary": 45000,
  "city": 'Pune',
  "joining_date": 2024-11-15T00:00:00.000Z
}
{
  "_id": ObjectId('69a14630d4aece84a622d794'),
  "eid": '003',
  "name": 'Amit Sharma',
  "age": 28,
  "department": 'HR',
  "designation": 'HR Executive',
  "salary": 45000,
  "city": 'Pune',
  "joining_date": 2024-11-15T00:00:00.000Z
}
> db.employees.countDocuments({ department: "IT" })
< 4
companydb>
```

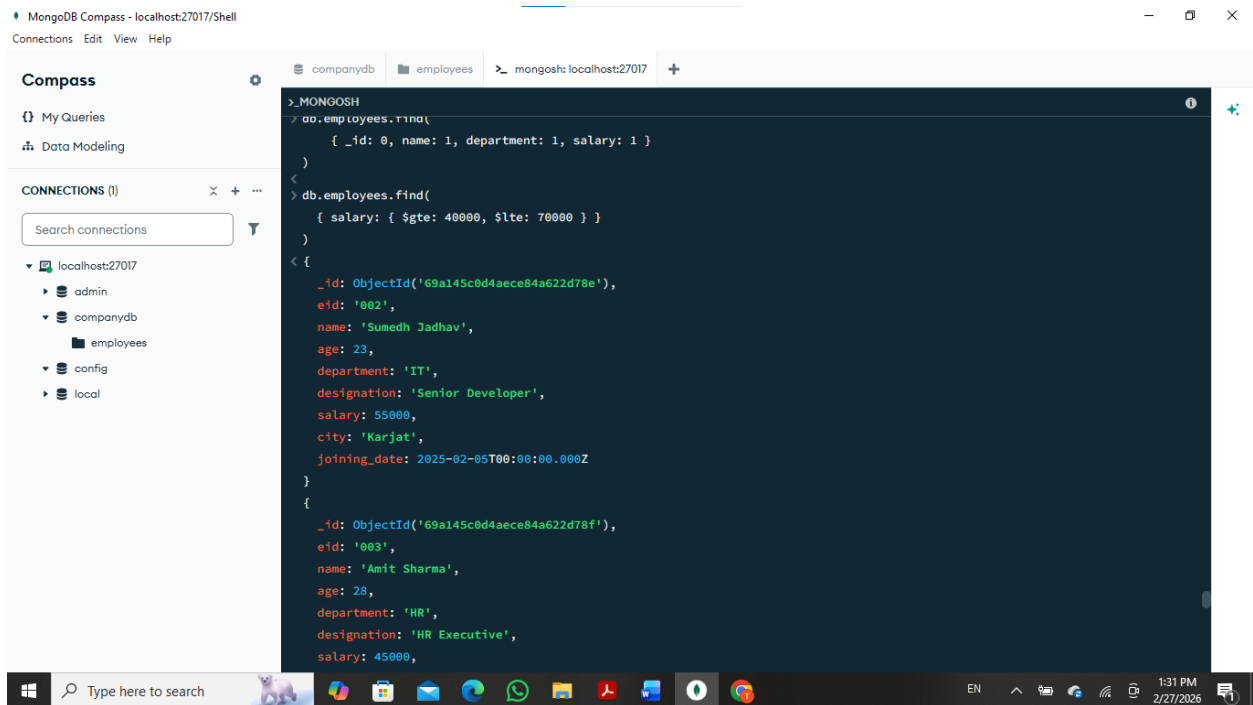
Display only "name", "department", and "salary" fields (hide _id).



The screenshot shows the MongoDB Compass interface. The left sidebar displays the 'CONNECTIONS' list with 'localhost:27017' expanded, showing databases like 'companydb' and 'employees'. The main workspace is in 'Shell' mode, showing a MongoDB query and its results.

```
>_MONGOSH
salary: 45000,
city: 'Pune',
joining_date: 2024-11-15T00:00:00.000Z
}
> db.employees.countDocuments({ department: "IT" })
< 4
> db.employees.find(
  {},
  { _id: 0, name: 1, department: 1, salary: 1 }
)
< [
  {
    name: 'Sumedh Jadhav',
    department: 'IT',
    salary: 55000
  },
  {
    name: 'Amit Sharma',
    department: 'HR',
    salary: 45000
  },
  {
    name: 'Priya Deshmukh',
    department: 'Finance',
    salary: 50000
  }
]
```

Find employees with salary between 40,000 and 70,000.



The screenshot shows the MongoDB Compass interface. The left sidebar displays the 'CONNECTIONS' list with 'localhost:27017' expanded, showing databases like 'companydb' and 'employees'. The main workspace is in 'Shell' mode, showing a MongoDB query and its results.

```
>_MONGOSH
> db.employees.find(
  { _id: 0, name: 1, department: 1, salary: 1 }
)
<
> db.employees.find(
  { salary: { $gte: 40000, $lte: 70000 } }
)
< [
  {
    _id: ObjectId('69a145c0d4aece84a622d78e'),
    eid: '002',
    name: 'Sumedh Jadhav',
    age: 23,
    department: 'IT',
    designation: 'Senior Developer',
    salary: 55000,
    city: 'Karjat',
    joining_date: 2025-02-05T00:00:00.000Z
  },
  {
    _id: ObjectId('69a145c0d4aece84a622d78f'),
    eid: '003',
    name: 'Amit Sharma',
    age: 28,
    department: 'HR',
    designation: 'HR Executive',
    salary: 45000,
  }
]
```

Find employees who joined after 2023-01-01.

MongoDB Compass - localhost:27017/Shell

Connections Edit View Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

localhost:27017

- admin
- companydb
 - employees
- config
- local

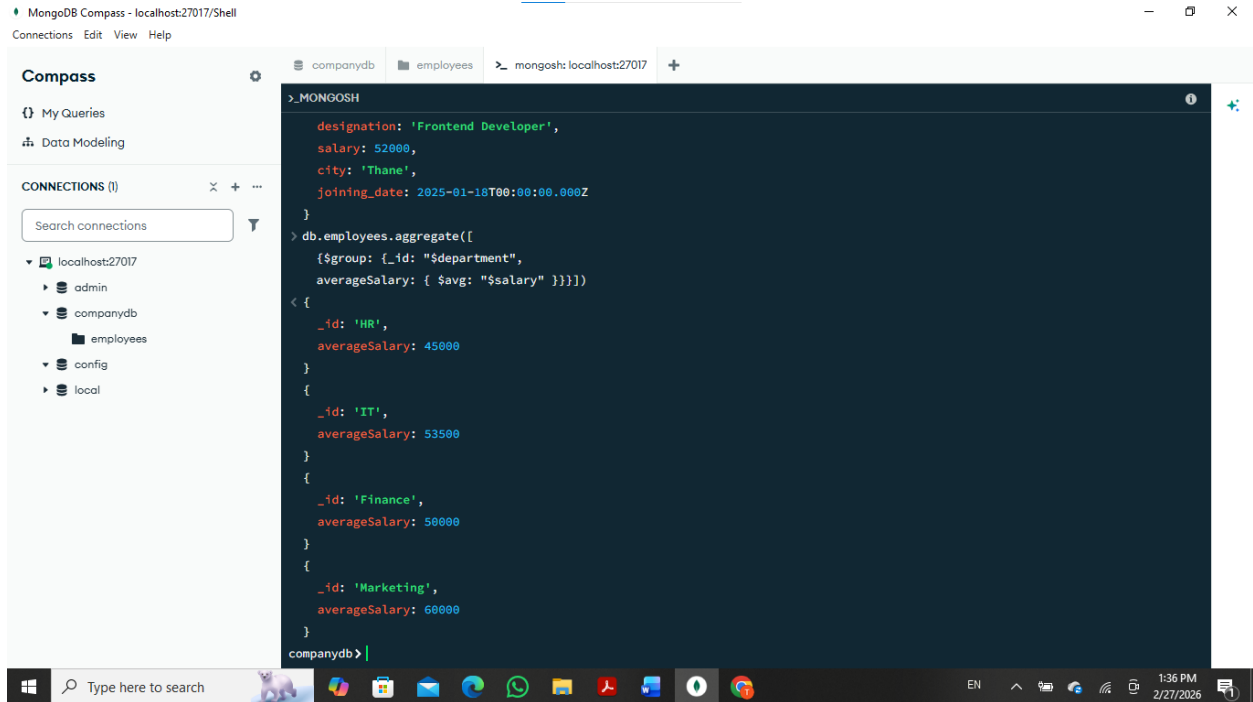
companydb employees mongosh: localhost:27017

```
>_MONGOSH
>use companydb
>use employees
>db.employees.find(
  { joining_date: { $gt: new Date("2023-01-01") } }
)
< {
  _id: ObjectId('69a145c0d4aece84a622d78e'),
  eid: '002',
  name: 'Sumedh Jadhav',
  age: 23,
  department: 'IT',
  designation: 'Senior Developer',
  salary: 55000,
  city: 'Karjat',
  joining_date: 2025-02-05T00:00:00.000Z
}
{
  _id: ObjectId('69a145c0d4aece84a622d78f'),
  eid: '003',
  name: 'Amit Sharma',
  age: 28,
  department: 'HR',
  designation: 'HR Executive',
  salary: 45000,
  city: 'Pune',
  joining_date: 2024-11-15T00:00:00.000Z
}
```

1:34 PM 2/27/2026

Task 6: Aggregation

Group employees by department and calculate the average salary.

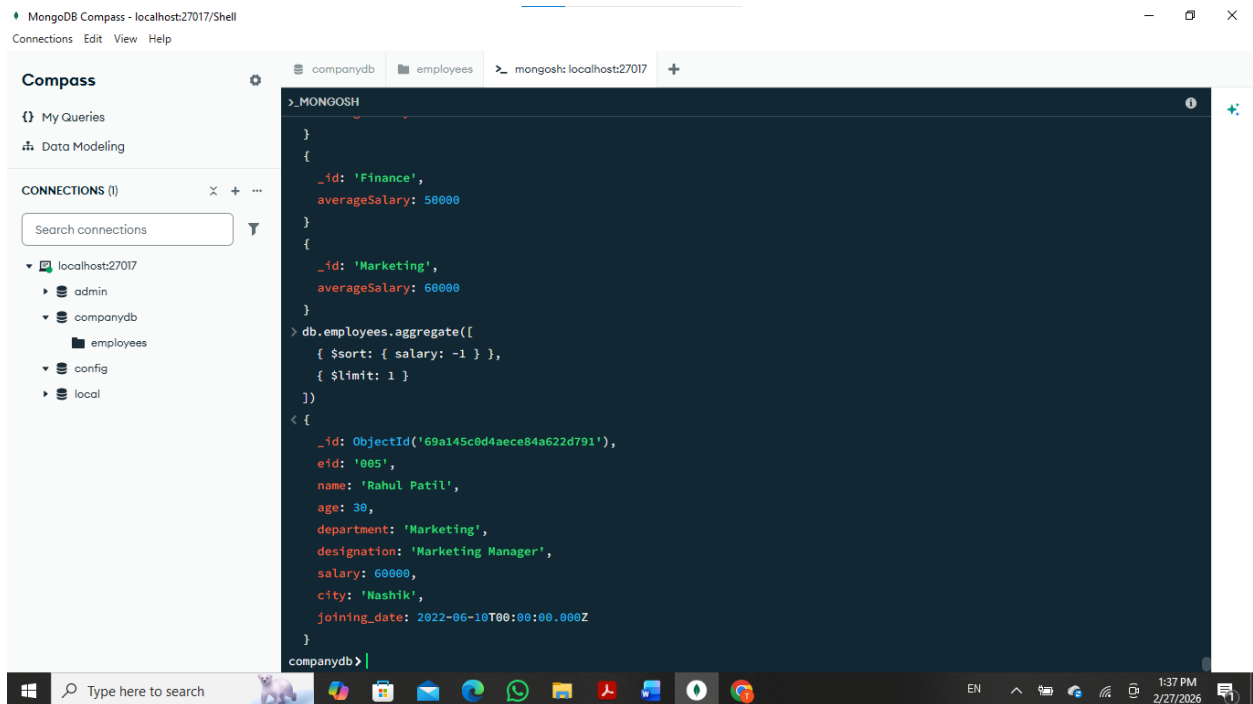


The screenshot shows the MongoDB Compass interface. The left sidebar displays the 'Connections' panel with a search bar and a list of connections. The main panel shows the 'companydb' database with the 'employees' collection selected. The 'MongoShell' tab is active, displaying the following aggregation query and its results:

```
>_MONGOSH
designigation: 'Frontend Developer',
salary: 52000,
city: 'Thane',
joining_date: 2025-01-18T00:00:00.000Z
}
> db.employees.aggregate([
  {$group: {_id: "$department",
    averageSalary: { $avg: "$salary" }}}])
< {
  _id: 'HR',
  averageSalary: 45000
}
{
  _id: 'IT',
  averageSalary: 53500
}
{
  _id: 'Finance',
  averageSalary: 50000
}
{
  _id: 'Marketing',
  averageSalary: 60000
}
}
companydb>
```

The results show the average salary for each department: HR (45000), IT (53500), Finance (50000), and Marketing (60000).

Find the highest paid employee in the company.



The screenshot shows the MongoDB Compass interface. The left sidebar displays the 'Connections' panel with a search bar and a list of connections. The main panel shows the 'companydb' database with the 'employees' collection selected. The 'MongoShell' tab is active, displaying the following aggregation query and its results:

```
>_MONGOSH
}
{
  _id: 'Finance',
  averageSalary: 50000
}
{
  _id: 'Marketing',
  averageSalary: 60000
}
> db.employees.aggregate([
  { $sort: { salary: -1 } },
  { $limit: 1 }
])
< {
  _id: ObjectId('69a145c0d4aeca84a622d791'),
  eid: '005',
  name: 'Rahul Patil',
  age: 30,
  department: 'Marketing',
  designation: 'Marketing Manager',
  salary: 60000,
  city: 'Nashik',
  joining_date: 2022-06-10T00:00:00.000Z
}
companydb>
```

The results show the highest paid employee: Rahul Patil, Marketing Manager, with a salary of 60000.

Count the number of employees in each city.

MongoDB Compass - localhost:27017/Shell

Connections Edit View Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

localhost:27017

- admin
- companydb
 - employees
- config
- local

```
>_MONGOSH
age: 30,
department: 'Marketing',
designation: 'Marketing Manager',
salary: 69000,
city: 'Nashik',
joining_date: 2022-06-16T08:00:00.000Z
}
> db.employees.aggregate([
  {$group: {_id: "$city", totalEmployees: { $sum: 1 }}}])
< [
  {
    _id: 'Mumbai',
    totalEmployees: 2
  },
  {
    _id: 'Karjat',
    totalEmployees: 2
  },
  {
    _id: 'Pune',
    totalEmployees: 2
  },
  {
    _id: 'Thane',
    totalEmployees: 2
  }
]
```

Type here to search

1:38 PM 2/27/2026

Find the total salary expense for the "IT" department.

MongoDB Compass - localhost:27017/Shell

Connections Edit View Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

localhost:27017

- admin
- companydb
 - employees
- config
- local

```
>_MONGOSH
_id: 'Karjat',
totalEmployees: 2
}
{
  _id: 'Pune',
  totalEmployees: 2
}
{
  _id: 'Thane',
  totalEmployees: 2
}
{
  _id: 'Nashik',
  totalEmployees: 2
}
> db.employees.aggregate([{$match: { department: "IT" } },
  {$group: {
    _id: "$department",
    totalSalaryExpense: { $sum: "$salary" }
  }}])
< [
  {
    _id: 'IT',
    totalSalaryExpense: 214000
  }
]
```

companydb >

Type here to search

1:39 PM 2/27/2026

